A SEPARATED FINITE TYPE SCHEME OF DIMENSION AT MOST ONE WITH NO PROPER ONE-DIMENSIONAL COMPONENT IS AFFINE

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- Affineness criterion for a curve: a curve is affine unless it is proper of dimension one. Stacks Project, Varieties, Lemma lemma-curve-affine-projective (tag 0A28). Makes each one-dimensional component affine under the hypothesis of the lemma.
- Image of an affine scheme under a finite surjective morphism is affine. Stacks Project, Cohomology of Schemes, Lemma coherent-lemma-image-affine-finite-morphism-affine-Noetherian (tag 01YQ). Transfers affineness from the coproduct of the components to X.

Source reference. Original reference 01J4: sources/stacks-chow/tags.txt, source lines 2001–2001. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original lemma 0A27: sources/stacks-chow/tags.txt, source lines 12695–12695. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

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Lemma 0.1. *Let X be a separated scheme of finite type over k. If $\dim(X) \leq 1$ and no irreducible component of X is proper of dimension 1, then X is affine.*

Proof. Let $X = \bigcup X_i$ be the decomposition of X into irreducible components. We think of X_i as an integral scheme (using the reduced induced scheme structure, see Schemes, Definition 01J4). In particular X_i is a singleton (hence affine) or a curve hence affine by Lemma 0A27. Then $\coprod X_i \to X$ is finite surjective and $\coprod X_i$ is affine. Thus we see that X is affine by Cohomology of Schemes, Lemma 01YQ. □

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